



**University of
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MODULE TITLE: Secure Networks

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CET260 - Assignment Part 2

Secure Network Implementation - Commentary

Task 4 - Commentary on Security Features

1. Choice of IP Version

For this network I decided to use IPv4. The main reason is that IPv4 is what I am more familiar with and it works fine for the size of network we are building here. The assignment said to use the 10.0.0.0/8 range so I used that. IPv4 gives us more than enough addresses for both sites.

I know IPv6 is newer and has more addresses but for this kind of internal company network, IPv4 with private addressing is still very common and practical. I used subnets like 10.1.1.0/24 for the Branch LAN and 10.2.1.0/24 and 10.2.2.0/24 for the HQ LANs. The WAN links to the ISP used the addresses provided in the starter file (1.0.0.0/28 for Branch side and 3.0.0.0/28 for HQ side).

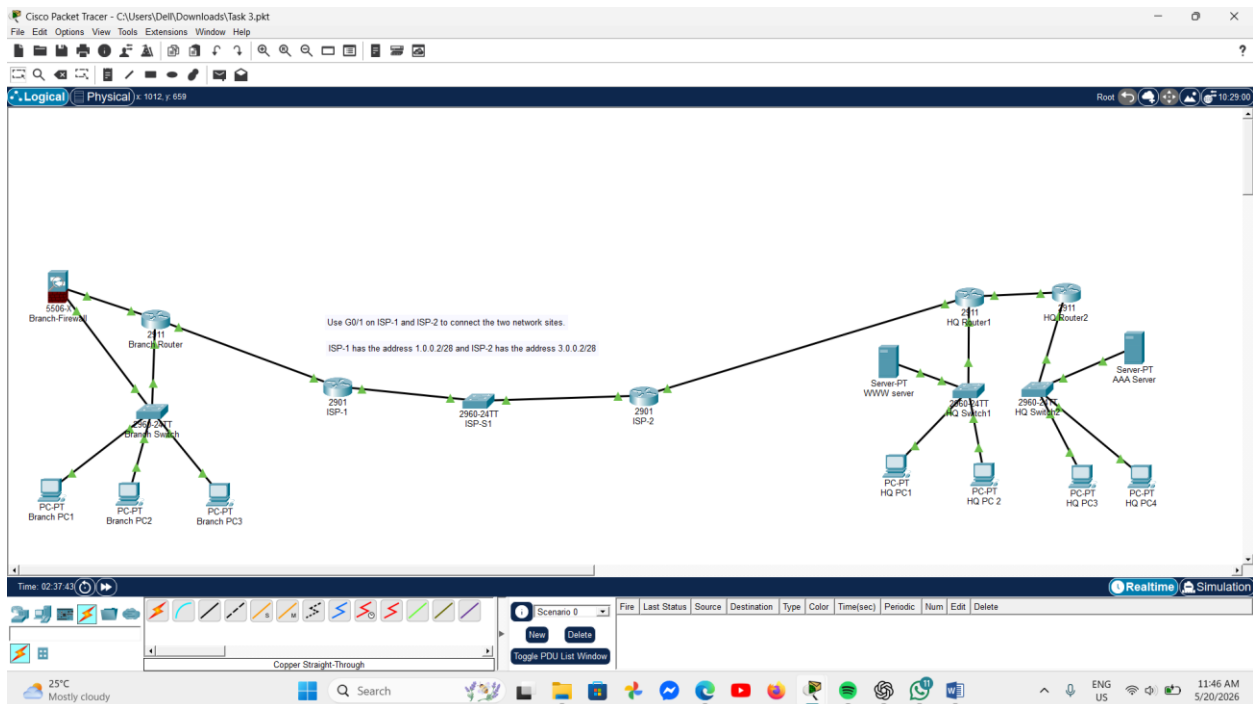


Figure 1: Complete network topology implemented in Cisco Packet Tracer

2. Implementation of Security Features

I implemented several security features across the network. Below I have explained each one.

Basic Device Security

On all three routers (Branch Router, HQ Router1 and HQ Router2) I configured passwords for both console access and remote access. I used 'enable secret' to protect privileged mode and also set a console password. I enabled SSH on all routers because Telnet sends passwords in plain text which is not safe. To set up SSH I had to configure a domain name and generate RSA keys with 1024 bits. I also enabled service password-encryption so that passwords stored in the config file are not visible in plain text.

```
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Branch-Router#
```

Figure 2: SSH configured on Branch Router

Firewall

I added a Cisco ASA 5506 firewall at the Branch site, placed between the Branch Router and the Branch Switch. The reason I put it there is so that all traffic coming in from outside has to pass through the firewall before it reaches the internal LAN. The inside interface was given security level 100 (most trusted) and the outside interface was given security level 0 (least trusted). The ASA automatically allows traffic from inside going out but blocks unsolicited traffic coming from outside. I also added an access list to allow traffic from the inside network to go out to the internet.

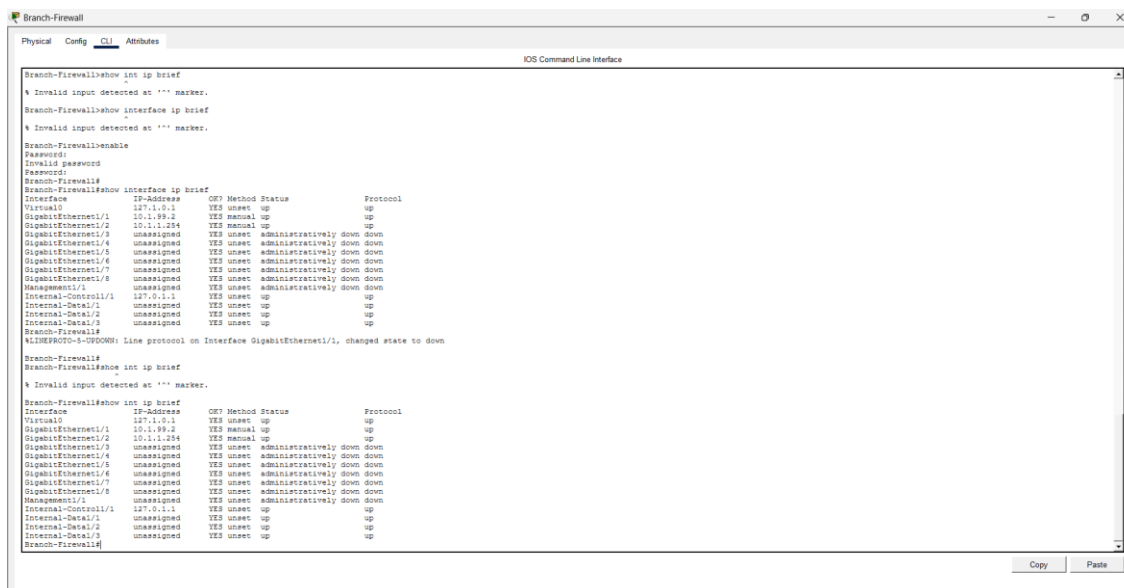


Figure 3: Firewall Interfaces

DHCP

I configured DHCP on both Branch Router and HQ Router1 so that PCs can get IP addresses automatically. I excluded the first 9 addresses in each subnet so that they are kept for routers and other devices that need static IPs. This is easier to manage than giving each PC a static address manually.

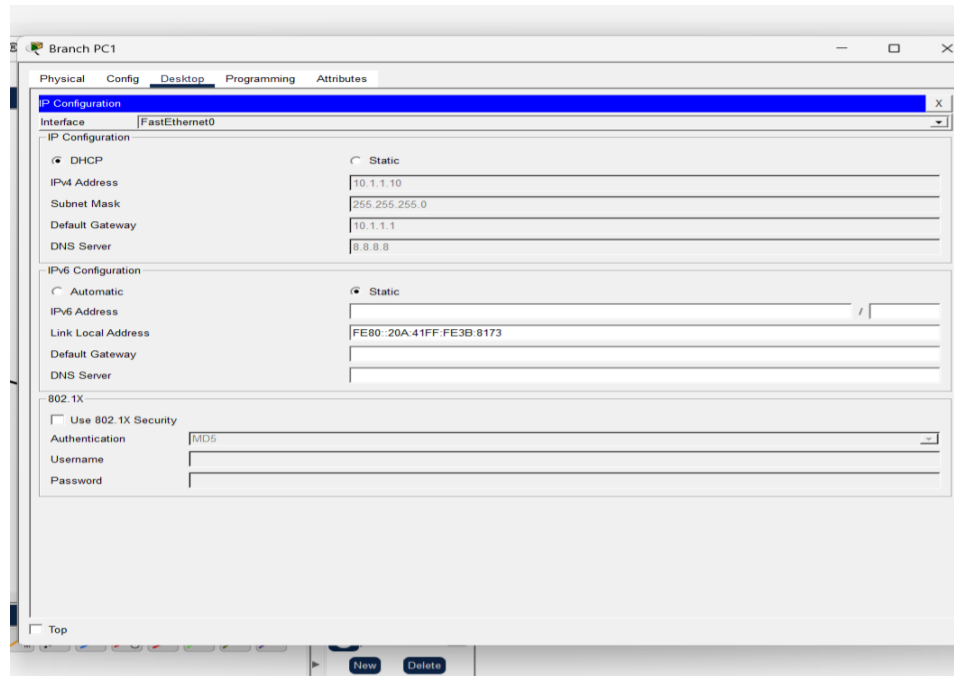


Figure 4: DHCP working on Branch PC1

OSPF Routing

I used static routing with default routes pointing to the ISP on both sides. This keeps the routing simple. OSPF was also considered but for a two-site network with relatively simple topology, static routes are more predictable and easier to manage.

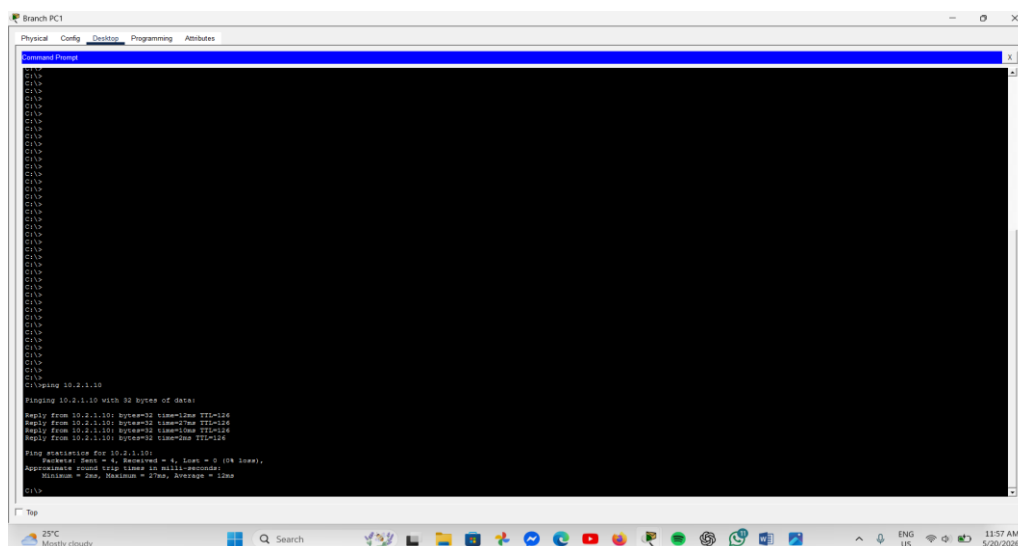


Figure 5: Ping test Branch to HQ

3. Reasoning Behind Firewall Type and NAT/PAT

I chose to use a stateful firewall (ASA) rather than a simple packet filter. A stateful firewall keeps track of connections so it knows which incoming packets are responses to requests made from inside the network. A packet filter just checks each packet individually without knowing the context, which is less secure.

I placed the firewall at the Branch site because that is where the main external connection is. For the HQ site I relied on the router ACLs and routing controls to manage traffic. In a more complete implementation I would add a firewall on the HQ side too but for this simulation I focused on the Branch side.

For NAT/PAT - I configured NAT on the Branch Router so that internal 10.x.x.x addresses get translated to the public IP when going to the internet. This is important because private IP addresses cannot be routed on the public internet. PAT (overloading) was used so that multiple internal devices can share one public IP address, which is how most real networks work.

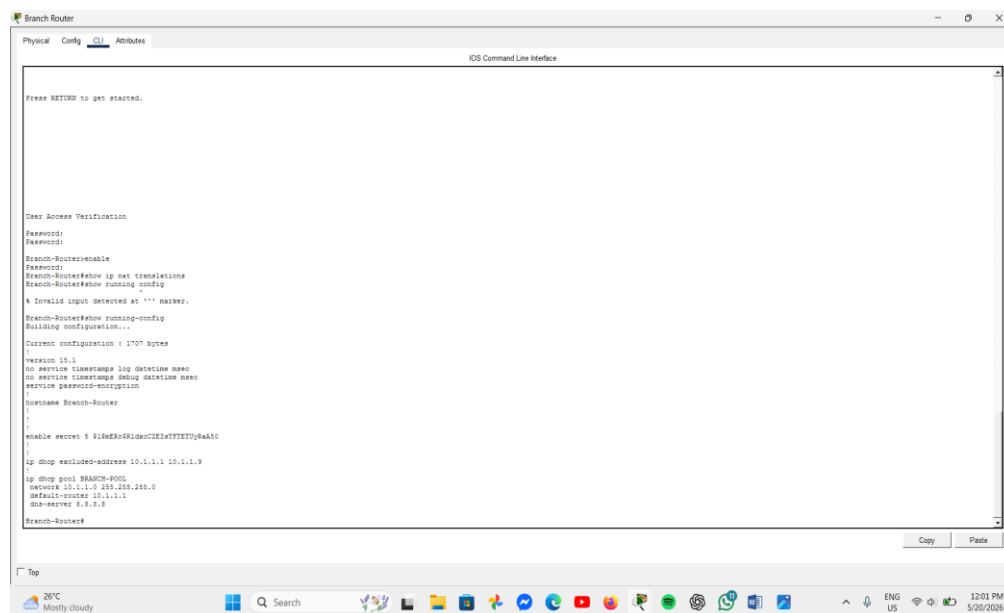


Figure 6: NAT/PAT configuration on Branch Router

4. VPN Parameters and GRE Tunnel

To connect the two sites securely I implemented a GRE tunnel between Branch Router and HQ Router1. The tunnel source is the WAN interface of Branch Router (1.0.0.1) and the destination is the WAN interface of HQ Router1 (3.0.0.1). The tunnel was assigned the IP address 10.9.9.1/30 on the Branch side and 10.9.9.2/30 on the HQ side.

GRE on its own does not encrypt traffic, it just wraps packets so they can travel across the internet. To add encryption I attempted to configure IPSec on top of the GRE tunnel using AES-256 encryption with SHA hashing and pre-shared keys. However Packet Tracer's simulated IOS version (15.1(4)M4) does not fully support the crypto isakmp commands needed for IPSec. In a real network deployment, IPSec would be configured with the following parameters:

- Encryption: AES-256 (strong encryption standard)

- Hashing: SHA (for integrity checking)
- Authentication: Pre-shared key
- DH Group: Group 2
- Lifetime: 86400 seconds (24 hours)

Even without IPSec the GRE tunnel was tested and working successfully - Branch PC1 was able to ping HQ PC1 (10.2.1.10) through the tunnel, which shows the inter-site connectivity is working.

```
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Branch-Router#
Branch-Router#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
Hardware is Tunnel
Internet address is 10.9.9.1/30
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 1.0.0.1 (GigabitEthernet0/1), destination 3.0.0.1
Tunnel protocol/transport GRE/IP
  Key disabled, sequencing disabled
  Checksumming of packets disabled
Tunnel TTL 255
Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (Kbps)
Tunnel receive bandwidth 8000 (Kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
16 packets input, 2048 bytes, 0 no buffer
Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 output buffer failures, 0 output buffers swapped out
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#
```

Figure 7: GRE Tunnel up

```
Branch-Router#
Branch-Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 1.0.0.2 to network 0.0.0.0

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/28 is directly connected, GigabitEthernet0/1
L    1.0.0.1/32 is directly connected, GigabitEthernet0/1
S    3.0.0.0/28 is subnetted, 1 subnets
     3.0.0.0/28 [1/0] via 1.0.0.2
S    10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
C    10.1.99.0/30 is directly connected, GigabitEthernet0/2
L    10.1.99.1/32 is directly connected, GigabitEthernet0/2
S    10.2.0.0/16 [1/0] via 10.9.9.2
C    10.9.9.0/30 is directly connected, Tunnel0
L    10.9.9.1/32 is directly connected, Tunnel0
S*   0.0.0.0/0 [1/0] via 1.0.0.2
Branch-Router#
Branch-Router#
```

Figure 8: tunnel Route in routing Table

Task 5 - Security Policy Document Extract

The following are five security policy rules that I would recommend for the Carlisle company network to help maintain stable and consistent security across both sites.

Rule 1 - Password Policy

All network devices including routers, switches and firewalls must have strong passwords configured. Passwords must be at least 12 characters long and must include a mix of uppercase letters, lowercase letters, numbers and special characters. Passwords must be changed every 90 days. Default or blank passwords are not allowed on any device. All passwords stored in device configurations must be encrypted using the service password-encryption command or equivalent. The reason for this rule is that weak passwords are one of the most common ways attackers gain access to network devices. A strong password policy makes it much harder for unauthorised people to log in.

Rule 2 - Remote Access Policy

Remote management of all network devices must only be done using SSH (Secure Shell). The use of Telnet is strictly prohibited on all devices because Telnet sends all data including usernames and passwords in plain text, which means anyone monitoring the network traffic could see them. SSH encrypts the session so login credentials and commands cannot be intercepted. All VTY lines on routers and switches must be configured with transport input ssh only. Additionally only authorised IT staff should have SSH access and their access should be logged.

Rule 3 - Firewall and Access Control Policy

A firewall must be deployed at the boundary between the internal network and the internet at both the Branch and HQ sites. The firewall must use a default deny policy, meaning all inbound traffic from outside is blocked unless it has been explicitly permitted by an access control rule. Outbound traffic from inside the network is permitted by default. Any exceptions or new rules must be documented and approved by the IT manager before being applied. The reason for this rule is that without a firewall any external user could attempt to connect to internal systems. A default deny approach means the company is protected from unknown or unexpected traffic.

Rule 4 - VPN and Inter-Site Communication Policy

All network traffic that travels between the Carlisle Branch site and the Carlisle HQ site must be sent through the encrypted VPN tunnel. Direct unencrypted communication between the two sites over the internet is not allowed. The VPN must use IPSec with AES-256 encryption as a minimum standard. The GRE tunnel provides the connection and IPSec provides the encryption. VPN keys and certificates must be reviewed and renewed every year. The reason for this rule is that traffic travelling across the public internet can potentially be intercepted by third parties. Encrypting all inter-site traffic means that even if someone did capture the traffic, they would not be able to read it.

Rule 5 - Device Hardening Policy

All network devices must be hardened before being connected to the live network. Hardening means disabling any services or features that are not needed. This includes disabling CDP (Cisco Discovery Protocol) on external-facing interfaces as it can reveal device information to attackers, shutting down any unused router or switch interfaces using the shutdown command, disabling the HTTP server on routers if web management is not required, and removing any default usernames or demo accounts from devices. The reason for this rule is that every unnecessary service or open port is a potential entry point for an attacker. By reducing the attack surface of each device we make the overall network more secure.

Conclusion

In conclusion, the network is successfully designed and implemented in Cisco Packet Tracer for both the Carlisle Branch and HQ sites. The final topology includes routers, switches, servers and end devices are connected using IPv4 addressing and static routing. Internal security features such as SSH, password encryption, NAT/PAT and firewall protection are configured to improve the security of the network. A GRE tunnel was also implemented to allow communication between both sites across the simulated internet connection. The overall implementation provided stable connectivity and followed important network security practices suitable for a small business environment.